University of Yaoundé I
Department of Physics, Faculty of Science
P.O. Box 812, Yaoundé, Cameroon
myke-vital.sao@facsciences-uy1.cm

July 25, 2026

The Editors-in-Chief
Journal of Cheminformatics
BioMed Central
London, UK

Dear Editors,

We are pleased to submit our manuscript, **"Persistent homology decomposes the scaffold paradox in AI-generated African antimalarial candidates"**, for consideration as an **Original Research Article** in the *Journal of Cheminformatics*.

Scientific contribution. This work benchmarks three quantum-inspired molecular representations—Topological Fingerprint (TFP), Tensor Network Embedding (TNE), and Quantum Kernel Score (QKS)—against classical fingerprints on a library of 19 849 African natural product-derived antimalarial candidates. We make five specific contributions of interest to the *Journal of Cheminformatics* readership:

1. **Systematic TDA–tensor–quantum benchmark on African natural product space.** We apply persistent homology (TDA), Tucker decomposition (TNE), and quantum circuit-based kernels (QKS) to the same chemical library under identical cross-validation conditions, enabling direct comparison with ECFP4, MACCS, FCFP4, AP, PHCO, and BPF baselines. Classical ECFP4 (AUC 0.868) remains the recommended primary screening baseline; the hybrid descriptor (AUC 0.842, $p = 0.111$) matches it without significantly exceeding it. An apparent quantum advantage in an early benchmark (QK AUC 0.936 vs. RBF 0.105) is shown to be an artefact of untuned RBF hyperparameters; after proper gamma optimisation, all kernels are statistically indistinguishable ($p > 0.05$). We report these negative results prominently, consistent with the open science values of the *Journal of Cheminformatics*.
2. **SOTA topological descriptor benchmark (new).** We benchmark five persistence-based descriptor strategies (PersStats, TFP-12, PersImage, BettiCurve, TFP-Enriched) with both Random Forest and SVM classifiers on the full 19 849-molecule library (RF) and a stratified subsample of $n = 5000$ (SVM, due to $O(N^2)$ scaling). PersStats + RF achieves AUC 0.873 with only 22 topological features, numerically matching the ECFP4 baseline (AUC 0.868) — among the closest topological approaches to classical fingerprints reported to date. RF consistently outperforms SVM across all strategies (mean ΔAUC = +0.071), establishing that tree-based methods better capture TDA feature interactions for this chemical space.
3. **Resolution of the scaffold paradox via persistent homology.** We demonstrate mathematically why a generative AI model simultaneously achieves 92.6% Tanimoto novelty and 69.3% scaffold recovery—a counterintuitive result that classical fingerprints cannot explain. Persistent homology decomposes this into H_1 ring topology (scaffold identity, preserved) versus H_0 connectivity (atom arrangement, divergent), providing a systematic topological resolution of a chemical space data inconsistency.
4. **Practical compression for resource-limited settings.** TNE compresses molecular tensors to a 192-element core descriptor at 5.9× real-atom compression, processed in under 30

minutes on a standard workstation. This has direct practical value for African and other resource-constrained research settings.

5. **Methodological discovery: balanced-class sampling for topological predictor validation.** A cross-paper analysis linking H_1 topological persistence with resistance resilience scores (RRS) showed a strong pilot correlation ($\rho = 0.947$, $n = 14$) that attenuated but remained significant upon expansion to 77 compounds ($\rho = 0.312$, $p = 0.006$ for H_1 count). Rather than interpreting this as a failed replication, we identify it as a methodological finding: small, class-imbalanced cohorts inflate apparent effect sizes, and balanced resistance-class sampling is essential for validating topological predictors of resistance tolerance. This is, to our knowledge, the first empirical identification of this confound in topological cheminformatics. Expanded ChEMBL search of 77 RRS-TFP compounds against 3 *Plasmodium* targets (231 pairs, Tanimoto ≥ 0.25) identified 7 structural analogues (3.0% match rate), of which 3 PfATP4 matches are experimentally active (IC_{50} 0.40–0.79 μM), while the remaining PfCRT matches are inactive; no PfDHFR analogues were found. The low match rate provides preliminary evidence of structural novelty; a TopologyNet analog (MLP on PersStats features, AUC 0.799 vs. RF AUC 0.860) confirms that tree-based methods better capture persistent homology feature interactions than neural architectures.

Open science and reproducibility. All TFP matrices, TNE embeddings, quantum kernel matrices, benchmark CSV files, and analysis scripts are deposited on Zenodo (DOI reserved: `https://doi.org/10.5281/zenodo.19608875`; data upload finalised upon acceptance) and mirrored on GitHub (`https://github.com/NanaEngo/Malaria_codesV2`) under the MIT licence. The computational protocol is fully reproducible on a standard CPU workstation with the `malaria_md` conda environment.

Fit with *Journal of Cheminformatics*. This manuscript directly addresses the journal's core scope: new molecular representation methods, rigorous benchmarking with classical baselines, applicability domain analysis, and open-source tool development. The African natural product focus aligns with the journal's equity and diversity mission. The honest reporting of negative results—including the debunking of an apparent quantum advantage, the demonstration that TFP alone does not outperform ECFP4, and the transparent attenuation of a cross-paper correlation—reflects the journal's commitment to scientific rigour. We position these quantum-inspired representations not as replacements for classical fingerprints, but as complementary diagnostic tools that reveal topological dimensions of molecular quality inaccessible to conventional cheminformatics.

Confirmation. This work is original, has not been published, and is not under consideration elsewhere. All authors have approved the submission. There are no conflicts of interest to declare.

We thank you for your consideration
and look forward to your response.

Myke-Vital Temgoua Sao
Corresponding Author

Suggested reviewers:

- Prof. Bing Wang (topological methods in drug discovery)

- Dr. Guillaume Godin (RDKit, natural product cheminformatics)
- Prof. Gisbert Schneider (generative molecular design)